

Experiment 6 : Vacuum Cleaner Problem (Python)

Aim

To write a Python program to simulate the **Vacuum Cleaner Problem**.

Algorithm

1. Start from the given room.
2. Check whether the room is dirty.
3. If dirty, clean the room.
4. Move to the next room.
5. Repeat until all rooms are clean.
6. Display the final state.

Program

```
rooms = {"A": "Dirty", "B": "Dirty"}
```

```
for room in rooms:
```

```
    print(room, ":", rooms[room])
```

```
    if rooms[room] == "Dirty":
```

```
        print("Cleaning", room)
```

```
        rooms[room] = "Clean"
```

```
print("\nFinal State")
```

```
for room in rooms:
```

```
    print(room, ":", rooms[room])
```

Input

Room A = Dirty

Room B = Dirty

Output

```
A : Dirty  
Cleaning A  
B : Dirty  
Cleaning B  
  
Final State  
A : Clean  
B : Clean
```

Result

Thus, the Python program to simulate the Vacuum Cleaner Problem was executed successfully, and all the dirty rooms were cleaned.